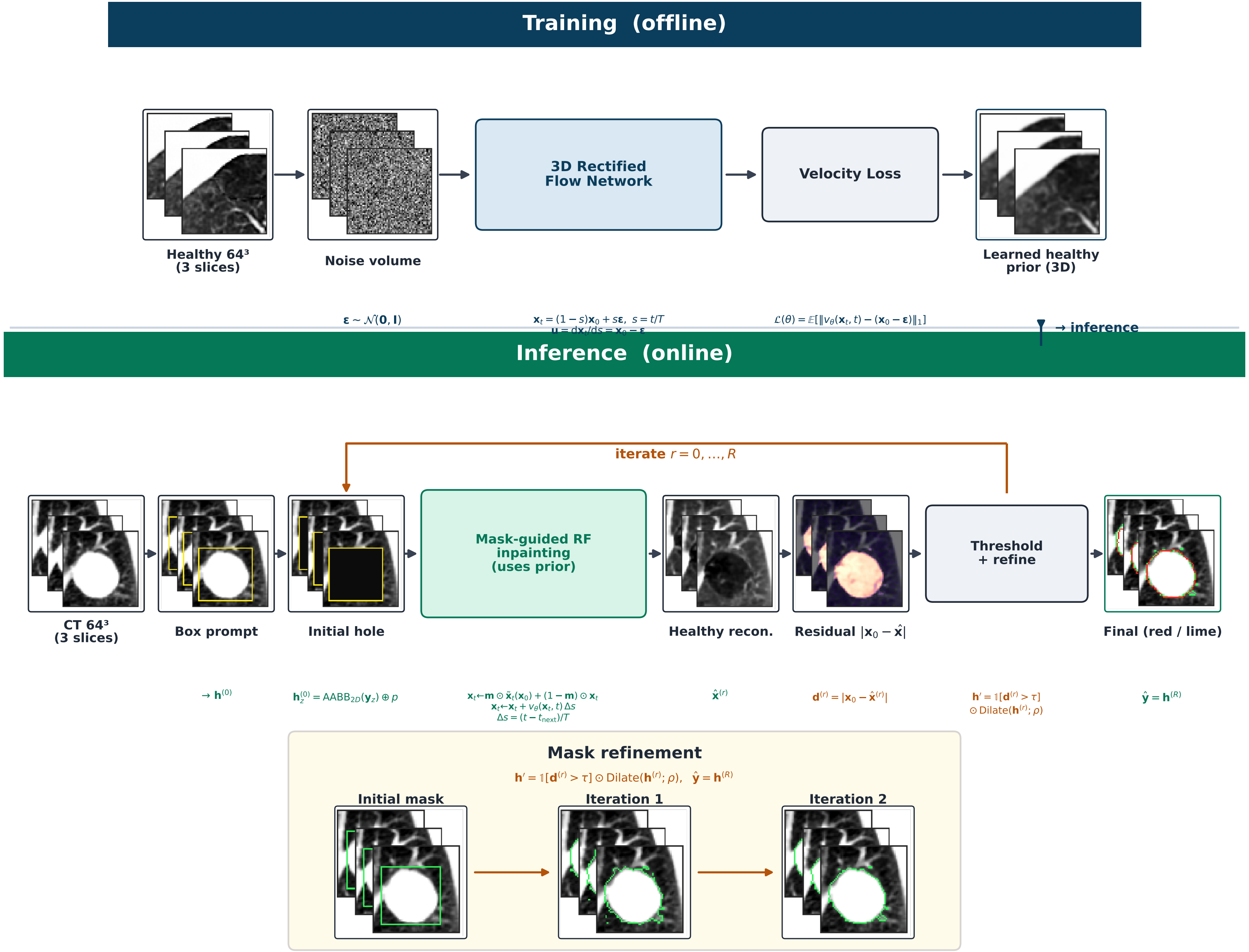

Training (offline)
Healthy 64³ (3 slices)
Noise volume
3D Rectified Flow Network
Velocity Loss
Learned healthy prior (3D)
$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\mathbf{x}_t = (1-s)\mathbf{x}_0 + s\boldsymbol{\varepsilon},\ s = t/T$
$\mathbf{u} = \mathrm{d}\mathbf{x}_t/\mathrm{d}s = \mathbf{x}_0 - \boldsymbol{\varepsilon}$
$\mathcal{L}(\theta) = \mathbb{E}[\|v_\theta(\mathbf{x}_t, t) - (\mathbf{x}_0 - \boldsymbol{\varepsilon})\|_1]$
→ inference
Inference (online)
iterate $r = 0, \ldots, R$
CT 64³ (3 slices)
Box prompt
Initial hole
Mask-guided RF inpainting (uses prior)
Healthy recon.
Residual $|\mathbf{x}_0 - \hat{\mathbf{x}}|$
Threshold + refine
Final (red / lime)
$\rightarrow \mathbf{h}^{(0)}$
$\mathbf{h}_z^{(0)} = \mathrm{AABB}_{2D}(\mathbf{y}_z) \oplus p$
$\mathbf{x}_t \leftarrow \mathbf{m} \odot \tilde{\mathbf{x}}_t(\mathbf{x}_0) + (1-\mathbf{m}) \odot \mathbf{x}_t$
$\mathbf{x}_t \leftarrow \mathbf{x}_t + v_\theta(\mathbf{x}_t, t)\,\Delta s$
$\Delta s = (t - t_{\text{next}})/T$
$\hat{\mathbf{x}}^{(r)}$
$\mathbf{d}^{(r)} = |\mathbf{x}_0 - \hat{\mathbf{x}}^{(r)}|$
$\mathbf{h}' = \mathbb{1}[\mathbf{d}^{(r)} > \tau]$
$\odot\, \mathrm{Dilate}(\mathbf{h}^{(r)}; \rho)$
$\hat{\mathbf{y}} = \mathbf{h}^{(R)}$
Mask refinement
$\mathbf{h}' = \mathbb{1}[\mathbf{d}^{(r)} > \tau] \odot \mathrm{Dilate}(\mathbf{h}^{(r)}; \rho),\quad \hat{\mathbf{y}} = \mathbf{h}^{(R)}$
Initial mask
Iteration 1
Iteration 2